

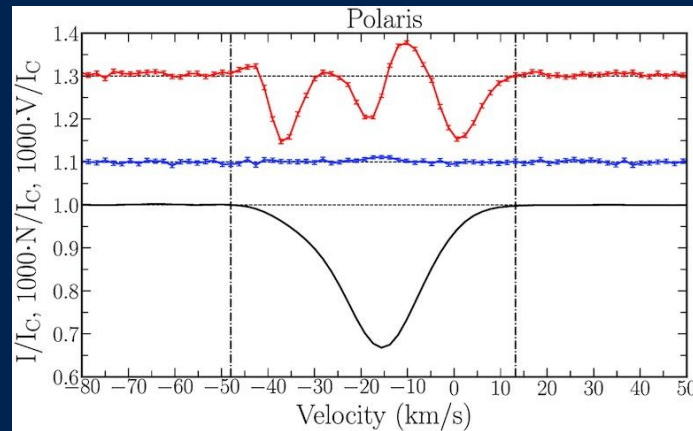
Stellar Magnetism in the Instability Strip: New Magnetic Detections in Classical and Type II Cepheids

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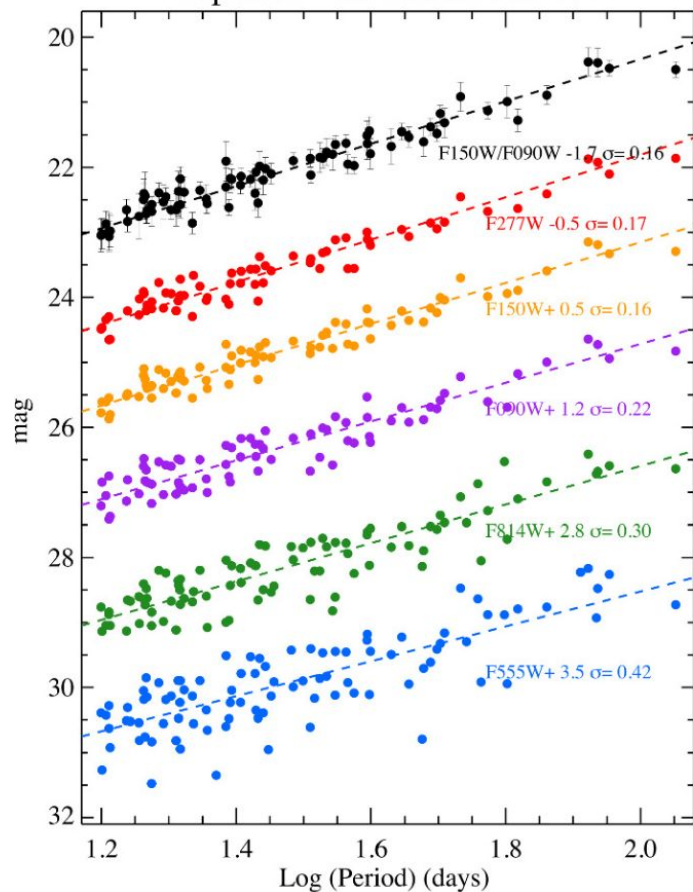
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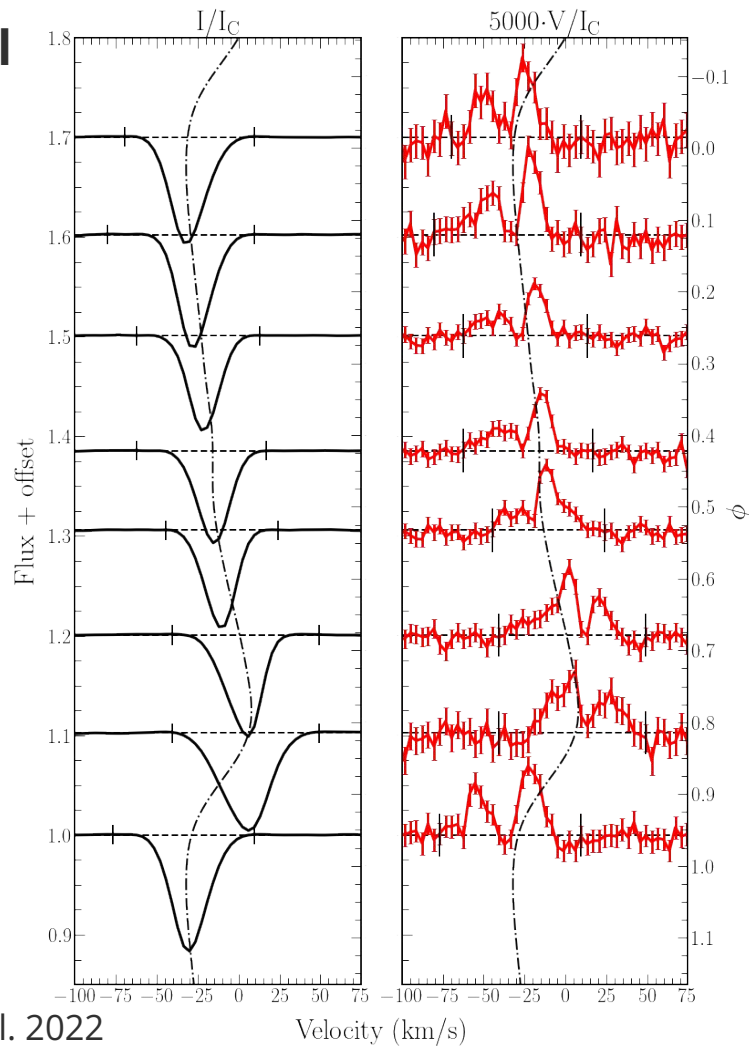
Introduction

Cepheid PL Relations in N1448



Riess et al. 2024

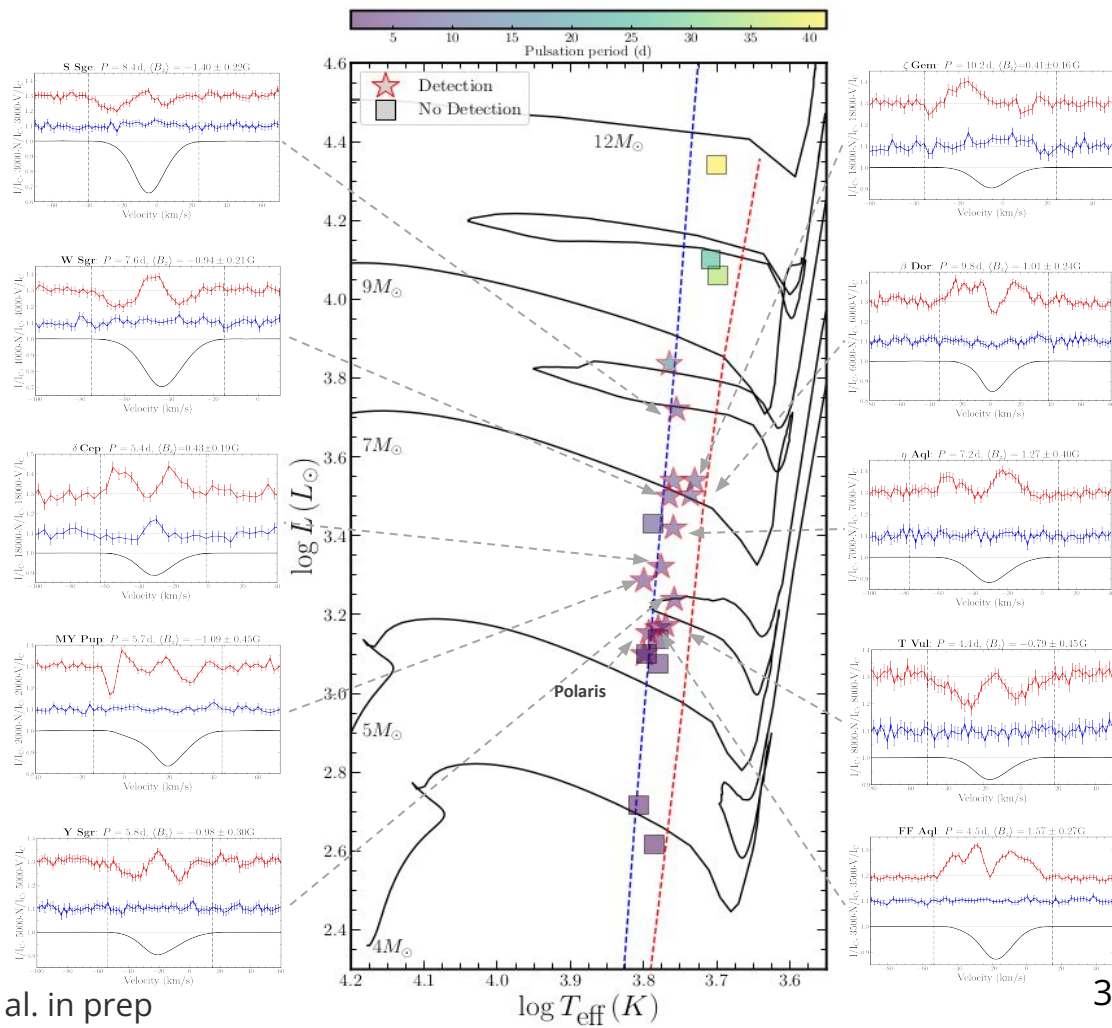
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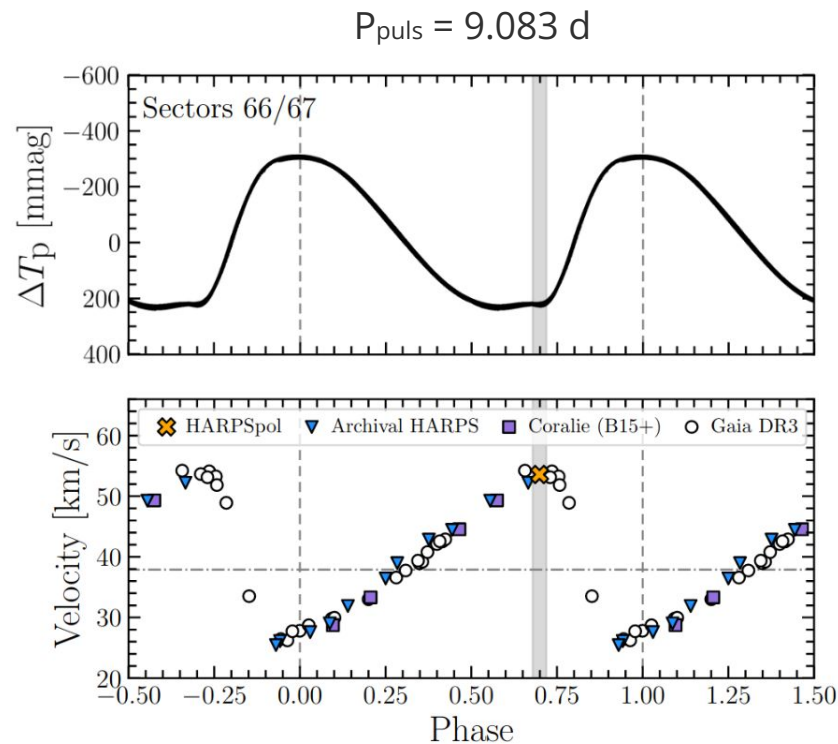
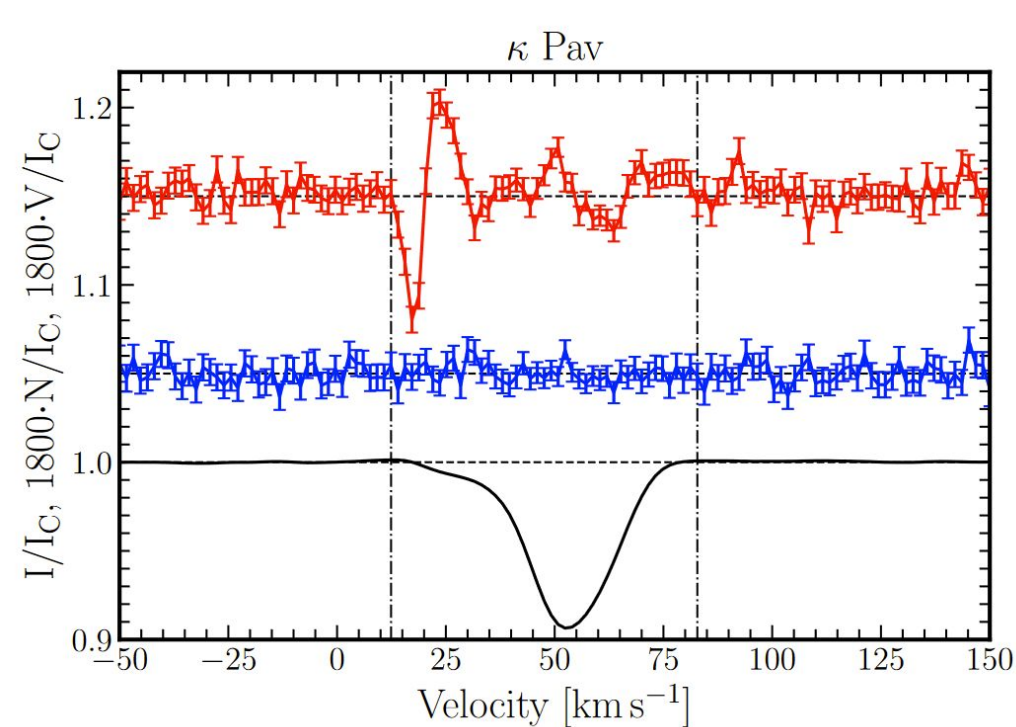
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Systematic survey of CCs

- Magnitude limited survey ($V < 6$ mag) of 26 of the brightest CCs using ESPaDOnS and HARPSpol.
- Observations obtained near photometric maximum.
- 14 total detections. 10/12 targets detected with ESPaDOnS when $S/N > 3000$ at 500 nm.
- Variety of asymmetric Stokes V morphologies which we attribute to the presence of atmospheric magnetic and velocity gradients.

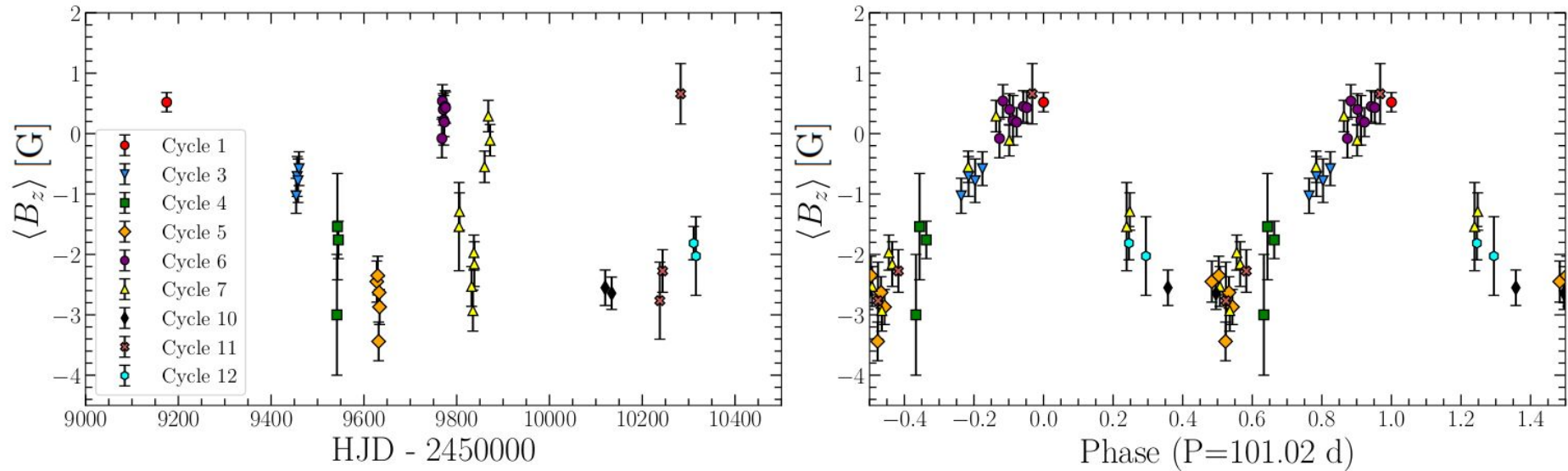


κ Pav: First magnetic detection in a W Vir Type II Cepheid



Barron et al. in prep

Polaris: Measuring rotation periods through spectropolarimetric monitoring



Conclusions

- Cepheid magnetic fields can be regularly detected at sub gauss precision when observed at ultra-high S/N using ESPaDOnS.
- Complex, asymmetric Stokes V morphologies indicate that circular polarization offers an exciting new probe of Cepheid atmospheric dynamics.
- Increased spectropolarimetric monitoring capabilities offered by Wenaokeao and Amakihi will allow for the direct measurement of rotation periods of evolved intermediate and massive stars.

